Measurable Is Not Reliance: Functional Shortcut Auditing in EEG Representations

Anonymous (FSR project draft)

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Abstract

Subject and domain information is strongly decodable from EEG representations, and it is often assumed that decodable leakage means the model relies on a harmful shortcut. We argue this inference is unlicensed: probing shows what is *encoded*, not what is *used*. We define a six-level functional-shortcut-reliance (FSR) ladder — detectability, reducibility, erasability, task coupling, functional reliance, and target consequence — and require any shortcut claim to relate levels rather than leap from a low one to a high one. We instantiate the ladder by re-analysing frozen artifacts across graph-CMI, concept-erasure, and branch-fusion pipelines, training nothing. First, measured leakage magnitude does not certify functional reliance: task-head alignment is positively associated with reliance while recomputable leakage carries the wrong sign. Second, subject signal is erasable, yet no eraser achieves a proven target benefit across 40 held-out-target cells, and aggressive erasure can harm the target. Third, a fusion backbone's spatial branch is load-bearing, but branch-local leakage and reliance are unmeasured and cannot be measured without a new run. FSR is an *audit framework*, not a domain-generalization method: it makes explicit what measurement is required to support what control conclusion.

1 Introduction

EEG representations carry strong subject- and recording-identity information: a linear or shallow probe recovers subject identity with high accuracy, and this signal is often entangled with the cohort/label structure of a dataset [Lin et al., 2026]. A common inference follows: if subject information is decodable from the representation, the model must be *relying* on it as a harmful shortcut, and removing it should improve cross-subject generalization. This paper argues that the inference is unlicensed and provides the audit that separates its premises from its conclusion.

The gap is the same one identified in the probing literature: a probe measures what is *encoded*, not what is *used* [Elazar et al., 2021]. Concept-erasure methods sharpen the point — they can remove a linearly-decodable attribute in closed form [Belrose et al., 2023] or by nullspace/adversarial projection [Ravfogel et al., 2020, 2022] — but removability is a statement about the representation, not a guarantee of a downstream benefit. And the domain- generalization literature warns that a moving invariance proxy does not imply a generalization gain once selection and baselines are controlled [Gulrajani and Lopez-Paz, 2021].

We formalize this as a six-level *functional-shortcut-reliance* (FSR) ladder (Figure 1): detectability (L1), reducibility (L2), erasability (L3), task coupling (L4), functional reliance (L5), and target consequence (L6). A harmful functional shortcut requires domain information that is simultaneously measurable, localizable, interventionable, functionally relied upon, and harmful in target consequence; a claim about a shortcut must therefore be a relationship *between* levels, never a leap from L1 to L5/L6.

Contributions. Re-analysing frozen artifacts (no new training) across a graph-CMI diagnostic, a frozen-feature concept-erasure study, and a branch-fusion backbone, we show: (1) **measurable $\neq$ relied-upon** — measured leakage magnitude does not certify functional reliance; in a frozen diagnostic, task-head alignment is positively associated with reliance while recomputable leakage carries the wrong sign; (2) **erasable $\neq$ beneficial** — subject signal is erasable, but erasure strength does not certify a target benefit (no eraser achieves a proven gain in 40 held-out-target cells); (3) **branch load matters but is unmeasured** — the

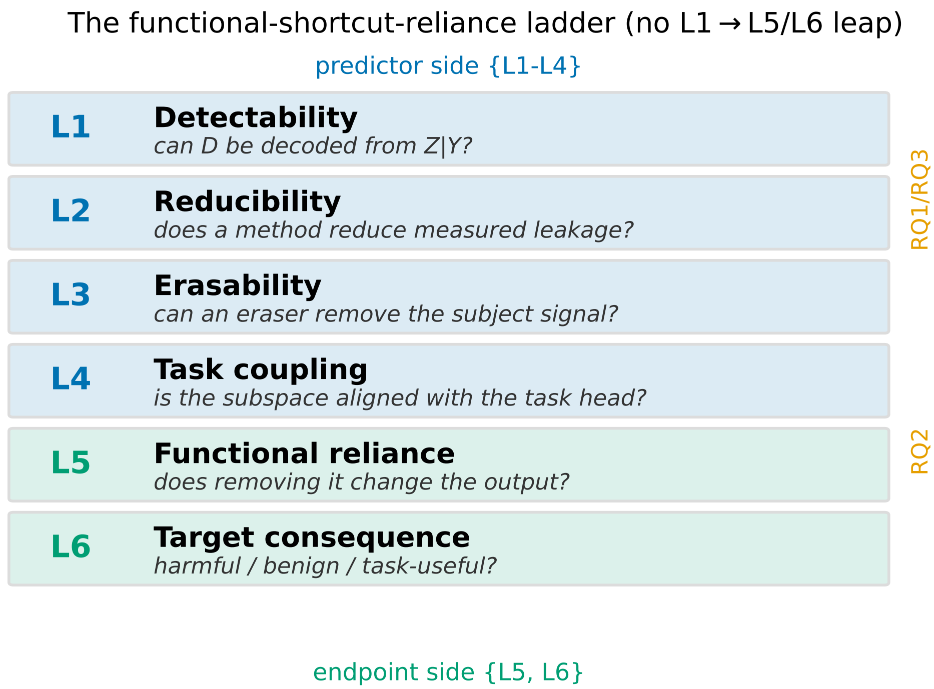

Figure 1: The six-level functional-shortcut-reliance ladder. Predictor-side levels {L1–L4} and endpoint-side levels {L5,L6}. A shortcut claim must relate levels; inferring L5/L6 from L1/L2 is forbidden.

spatial branch of a fusion backbone is load-bearing, yet branch-local leakage/reliance is not measured and cannot be without a new run. FSR is an audit framework, not a domain-generalization method; a companion premise is that conditional-invariance/CMI *control* on these representations is already a closed negative.

2 The FSR audit ladder

Let Z be an EEG representation, Y the task label, and D the domain (subject/recording). Figure 1 defines the six levels.

- **L1 Detectability.** Can D be decoded from Z given Y? A label-conditional domain probe / posterior-KL proxy $I_w(Z; D \mid Y)$ (a plug-in linear-probe surrogate, *not* an unbiased mutual information) with a within-label permutation null.
- **L2 Reducibility.** Does a method reduce the *measured* leakage (paired ΔKL / Δprobe accuracy vs a matched baseline)?
- **L3 Erasability.** Can a linear/non-linear eraser remove the subject signal? Post-eraser residual subject decodability for LEACE/INLP/RLACE/mean-scatter, with a same-rank random-k control.
- **L4 Task coupling.** Is the subspace aligned with the task head? The fraction of linear task-head row-space energy in the top-k label-conditional subject subspace (align_k), and, for multi-branch models, branch-ablation drop plus fusion-gate weight.
- **L5 Functional reliance.** Does removing the subspace change the task head's output? Head-replay task-drop $R3$ after removing the top-k subject subspace (source-only fit), with a random-subspace control and an exact-replay firewall.
- **L6 Target consequence.** Is that reliance harmful, benign, or useful? Held-out-target bAcc/NLL/ECE delta and worst-subject delta, with target labels used only for scoring.

Direction contract. Under the naive "leakage-is-shortcut" hypothesis, corr(leakage, $R3$) and corr(subject-removed, target-b
should be *positive*. The FSR questions ask whether they are.

Quantitative-inclusion gate. A route may enter a cross-level quantitative test only if it carries ≥ 1 predictor-side level {L1–L4} *and* ≥ 1 endpoint-side level {L5,L6}; otherwise it is tagged support/boundary/protocol/backgrou
and informs interpretation only. Of the audited routes, only a small set is quantitatively includable (Table 2).

3 Setup and provenance discipline

All results are re-analyses of *frozen* artifacts; we train, tune, and select nothing on the analysed data. Three pipelines supply the levels. **CIGL** (RQ1/RQ3): a static graph backbone on BCI-IV-2a (BNCI2014_001) and BNCI2015_001, leave-one-subject-out (LOSO), seeds {0,1,2}, providing per-unit align_k, graph leakage, and $R3$. **TOS** (RQ2): frozen-feature erasure on 2a, Lee2019, Cho2017, and Schirrmeister2017, backbones TSMNet [Kobler et al., 2022] (SPD, z=210) and EEGNet [Lawhern et al., 2018] (z=16), erasers LEACE [Belrose et al., 2023], INLP [Ravfogel et al., 2020], RLACE [Ravfogel et al., 2022], mean-scatter, and random-k; source-only fit, held-out-target deploy. **FBCSP-LGG** (RQ4): a filter-bank CSP [Ang et al., 2012] plus learnable-graph backbone with a three-way gated fusion over graph/temporal/spatial branches; branch ablation and gate weights only.

Claim-strength tiers. Every quantitative statement is tagged RECOMPUTED (recomputed from committed per-unit data), RECOMPUTED_SIGN_ONLY (only a recomputable slice confirms the sign), or FROZEN_NOT_RECOMPUTA
(carried from a frozen output whose per-unit inputs are unavailable — support only, never a reproduction). A fail-closed validator enforces the schema, and a reproduction script re-derives the headline and halts if it drifts beyond tolerance.

Target-label firewall. Target labels y_T are used only to score the L6 endpoint; they never enter probe/eraser fitting, model selection, hyper-parameter selection, or early stopping. Each route carries a fit-tag in {NO, YES_FORBIDDEN, AUDIT_ONLY, UNKNOWN}; every quantitative test uses only NO-tagged routes. One legacy route is YES_FORBIDDEN (a retracted, disclosed target-label leak) and is excluded from all tests (Table 2).

Statistics. Spearman correlations with a paired bootstrap CI (rng(0), 2000 resamples, percentile$[2.5, 97.5]$); the alignment-vs-leakage comparison uses a signed-difference bootstrap. Mechanism regressions use OLS with a z-scored outcome and predictors plus one-hot dataset/seed/method controls. Robustness includes dataset-stratification, per-seed, per-method, leave-one-dataset-out, within-group rank-residualization, and, for erasure, subset sensitivity across eraser families. All analyses are CPU-only over frozen artifacts.

4 Measurable is not reliance (RQ1/RQ3)

RQ1 — does measured leakage predict functional reliance? We report three claim-strength tiers. **(RQ1A, RECOMPUTED)** Task-head alignment is positively associated with $R3$: pooled Spearman ρ=+0.34 $[+0.17, +0.50]$, n=126; the association is significant on 2a ($\rho\approx$+0.47) but not on 2015 ($\rho\approx$+0.10), i.e. pooled-positive but not universal. **(RQ1B, RECOMPUTED_SIGN_ONLY)** Graph leakage carries the *wrong* sign: seed0 ρ=−0.42 $[-0.68, -0.10]$, n=42. **(RQ1C, FROZEN_NOT_RECOMPUTABLE)** The pooled leakage correlation −0.34 is carried from a frozen output whose per-fold inputs for two of three seeds were pruned; we do not present it as a reproduction. Thus measured leakage magnitude does not certify reliance, and where recomputable it points the wrong way.

RQ3 — is alignment a better reliance predictor than leakage? The primary test is the signed Spearman *difference* (align − leakage) = +0.816 $[+0.219, +1.333]$, which excludes zero: alignment is more positively associated with reliance than leakage. In a paired standardized regression on the recomputable

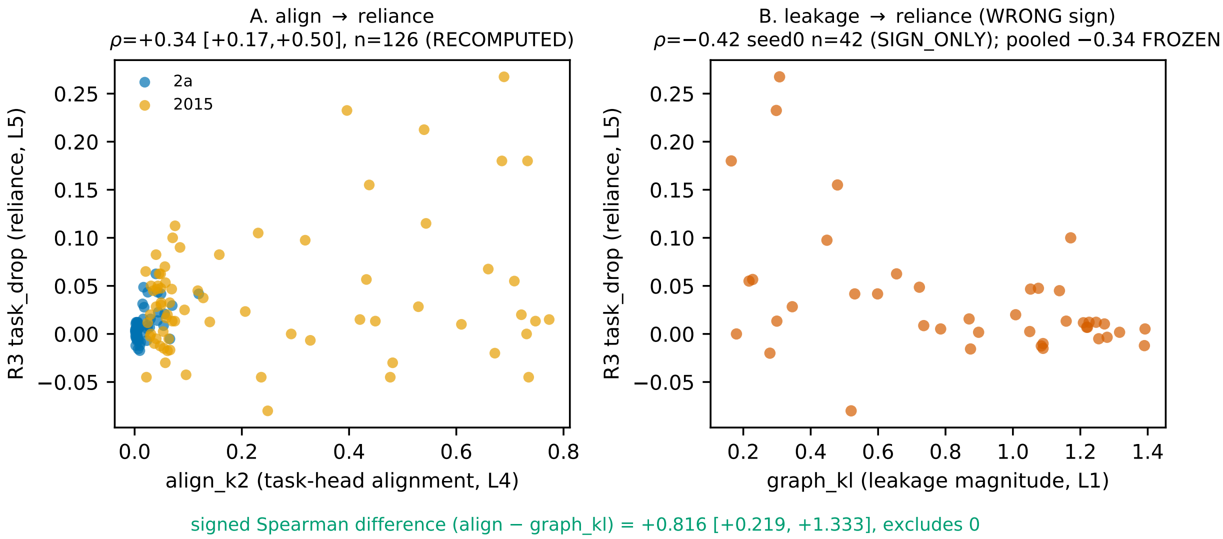

Figure 2: Alignment vs leakage as predictors of functional reliance ($R3$). **A**: task-head alignment is positively associated with reliance (ρ=+0.34 [+0.17, +0.50], n=126, RECOMPUTED); the association is dataset-heterogeneous (2a significant, 2015 not). **B**: recomputable graph leakage carries the wrong (negative) sign (ρ=−0.42 seed0, n=42, RECOMPUTED_SIGN_ONLY); the pooled −0.34 is FROZEN_NOT_RECOMPUTABLE. The signed Spearman difference (align − leakage) = +0.816 [+0.219, +1.333] excludes zero.

seed0 units, alignment is correctly signed (β=+0.26) and leakage wrongly signed (β=−0.37); with a dataset control, the alignment partial effect is positive but not individually significant, reflecting the RQ1A heterogeneity. We therefore state that *task-head alignment is closer to functional reliance than raw leakage in this frozen diagnostic* — and explicitly *not* that align_k is a validated reliance estimator.

5 Erasable is not beneficial (RQ2)

Subject signal is erasable: LEACE drives linear subject decodability to chance on both backbones (a nonlinear MLP residual persists), and INLP drives it to chance entirely. The question is whether that removability confers a target benefit.

No certified benefit. Deploying source-fitted erasers to held-out target subjects, we find **no eraser achieves a proven target-bAcc gain in any of 40 cells** (benefit_claimable = 0/40 under a rule requiring a positive point estimate whose 95% CI lower bound exceeds zero, no task-collapse, no binary-latent harm, and no random-k-matched non-specific NLL move; Figure 3A). Erasure can actively harm the target: INLP over-erases and drives task to chance in several cells, and on the compact binary EEGNet latent (Lee2019, Cho2017) LEACE and RLACE drive task-bAcc to chance, a task-safety failure. On 2a-TSMNet the small NLL improvement under LEACE is matched by a same-rank random removal that leaves the subject fully decodable, so that NLL movement is non-specific regularization, not domain removal.

The negative correlation is not a finding. Pooled across all 40 cells, ρ(subject-removed, Δtarget bAcc) is negative and excludes zero. A sensitivity battery (Figure 3B) shows this is not robust: it is non-significant when INLP or random-k are removed, and it *flips to* +0.54 (excluding zero) on the principled-eraser subset (LEACE/RLACE only). The negative sign is driven by INLP over-erasure and the random-k anchor, not by a "more removal, worse target" law. We therefore report only the robust result — the absence of a certified benefit — and do not headline a negative association. The upshot: subject signal is erasable, but erasure strength does not certify target benefit.

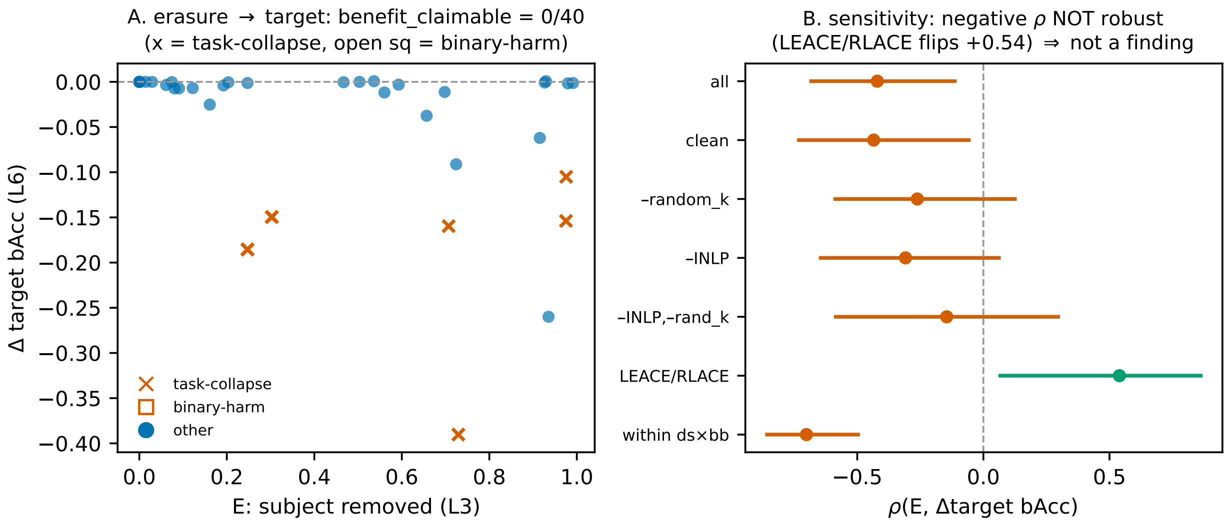

Figure 3: Erasure strength vs target benefit across 40 held-out-target cells. **A**: no eraser achieves a proven target-bAcc gain (benefit_claimable = 0/40); many cells task-collapse (INLP) or harm a binary-latent target (LEACE/RLACE on Lee/Cho EEGNet). **B**: the apparent negative ρ(subject-removed, Δtarget bAcc) is *not robust* — it flips to +0.54 on the principled-eraser subset (LEACE/RLACE) and is non-significant when INLP or random-k are dropped; it is therefore not reported as a finding.

6 Branch load matters, but branch-local shortcut is unmeasured (RQ4)

A multi-branch backbone lets us ask whether leakage *means* something different in a load-bearing branch than in a non-load-bearing one. In FBCSP-LGG the spatial branch is clearly load-bearing: zeroing it drops task-bAcc by 7.4 points on 2a and 8.8 on 2015, and it receives the largest fusion-gate weight (0.49/0.57); the graph and temporal branches are neutral-to-slightly-harmful and starved after fusion (Figure 4).

Blocked, not failed. Answering the branch-locality question quantitatively requires per-branch *leakage* (L1) and per-branch *reliance* (L5). Neither exists: there are no frozen per-branch latent dumps and no per-branch probe on disk, and — decisively — *no trained checkpoint exists at all.* The training pipeline persisted only scalar summaries; the best-epoch state was held in memory and never serialized, and a read-only search of the compute store returned zero matching checkpoints. Producing the metric would require retraining, which we do not do. RQ4 is therefore reported as blocked: we can state that the spatial branch is load-bearing, but we *cannot* say "spatial leakage is harmful," "graph leakage is benign," or "per-branch conditional MI predicts reliance." The blocked status is itself a finding: a branch-local shortcut claim requires an instrument that does not currently exist.

7 Related work

Probing measures encoding, not use. A probe establishes that an attribute is *decodable* from a representation, not that a model *uses* it; amnesic and counterfactual probing make the distinction operational by intervening on the representation and observing behavior, and show that decodability and causal use can dissociate [Elazar et al., 2021]. Our L1↔L5 separation is the EEG realization of this: a subject probe measures detectability, while head-replay $R3$ measures whether removing the subspace actually changes the task head's output.

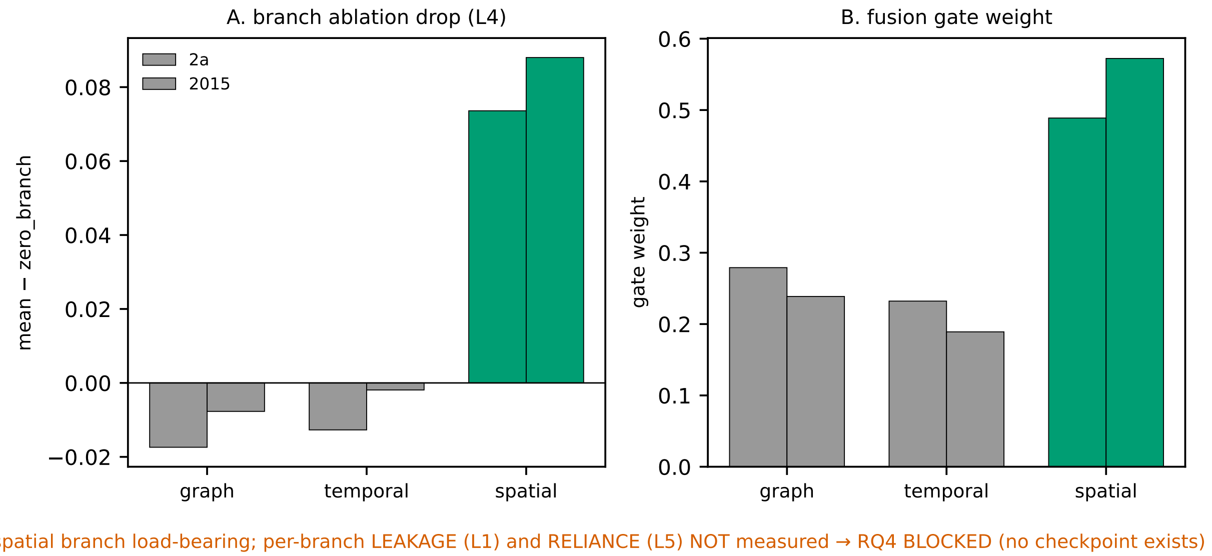

Figure 4: Branch-locality in the FBCSP-LGG fusion backbone. The spatial branch is load-bearing (largest ablation drop and highest fusion-gate weight on both datasets); graph/temporal branches are neutral-to-starved. Per-branch *leakage* (L1) and *reliance* (L5) are not measured, so RQ4 is blocked (no frozen checkpoint exists to measure them without retraining).

Concept erasure: removed is not beneficial. A family of methods removes a linearly-decodable concept from a representation: LEACE erases it in closed form with provably minimal collateral damage [Belrose et al., 2023]; INLP removes it by iterative nullspace projection [Ravfogel et al., 2020]; RLACE by a relaxed adversarial minimax [Ravfogel et al., 2022]; and scatter/moment methods target invariant subspaces through a generalized eigenproblem [Ghifary et al., 2017]. These are precisely our L3 operators. Prior work establishes *removability*; we deploy the same operators source-to-target and test the downstream consequence, finding that erasure strength does not certify a target benefit.

Domain generalization needs selection rigor. Under careful model selection with a strong ERM baseline, many invariance-based methods fail to beat ERM, and apparent gains often reflect selection rather than a robust effect [Gulrajani and Lopez-Paz, 2021]. This is the direct warning against reading a moving invariance proxy as a generalization gain, and it motivates our refusal to propose a new method: we audit whether the measured proxies even predict the reliance/consequence they are assumed to control.

EEG subject identity is strong and entangled. Audits of EEG foundation models report that subject identity is highly decodable and can confound label/cohort structure, an "identity trap" [Lin et al., 2026]. Because the signal is strong, equating "decodable subject" with "harmful shortcut" is tempting; our audit is designed to separate the two.

EEG decoders and branch structure. The backbones we analyse are standard: filter-bank common spatial patterns for oscillatory motor-imagery features [Ang et al., 2012], the compact convolutional EEGNet [Lawhern et al., 2018], and the SPD/tangent-space TSMNet with domain-specific normalization [Kobler et al., 2022]. Branch-locality (a gated fusion over graph/temporal/spatial branches) lets us ask whether leakage means something different in a load-bearing branch — a question we can pose but, for want of an instrument, not yet answer quantitatively.

Decompose a joint effect into pathways. A single unlabeled adaptation number conflates a geometry pathway and a prevalence/decision-prior pathway and admits an exact decomposition [Anonymous, 2026b].

FSR applies the same measurement-then-decompose discipline to a different object — leakage, erasure, task-coupling, functional reliance, and target consequence — rather than collapsing them into a single "shortcut" scalar.

8 Limitations

We state the limitations plainly; for an audit paper they are part of the contribution.

- **Provenance limit.** The pooled leakage→reliance correlation is not recomputable (per-fold leakage for two of three seeds was pruned); it is reproduced only at seed0 (sign) and carried as FROZEN_NOT_RECOMPUTABLE at pooled. Only alignment→reliance is fully recomputed.
- **Heterogeneity / small n.** The alignment→reliance association is significant on 2a but not on 2015; the alignment-vs-leakage contrast is at seed0, n=42, with non-significant within-group partial betas.
- align_k **is not a validated estimator**, only closer to reliance than raw leakage.
- **The RQ2 negative correlation is not a finding** — it is a non-robust confound of over-erasure and the random-k anchor (Figure 3B); the RQ2 result is the absence of a certified benefit (0/40).
- **RQ4 is blocked**, not answered: per-branch leakage/reliance are unmeasured and no checkpoint exists to measure them without retraining.
- **Proxies, not CMI.** L1/L2 use a label-conditional linear-probe posterior-KL surrogate and a linear-probe advantage, not an unbiased mutual information; we say "extractable conditional domain information," not "CMI."
- **Erasability $\neq$ full removal** (LEACE leaves a nonlinear residual); the one target-unlabeled positive we cite (a non-CMI adaptation control) is seed0-only support, not an FSR result.
- **Frozen scope.** All results re-analyse frozen artifacts on specific backbones; generality beyond them is not claimed.

9 Conclusion

We have argued, and shown on frozen EEG artifacts, that a decodable subject/domain signal is not by itself evidence of a harmful functional shortcut. Measurable leakage magnitude does not certify functional reliance — task-head alignment is closer to reliance, though not a validated estimator. Subject signal is erasable, but erasure strength does not certify a target benefit: no eraser achieves a proven gain across 40 held-out-target cells, and aggressive erasure can harm the target. Branch load matters — a fusion backbone's spatial branch is load-bearing — but branch-local leakage and reliance are unmeasured, and the instrument to measure them does not currently exist.

Measurable is not reliance. Erasable is not beneficial. Branch load matters, but branch-local shortcut claims require missing instruments.

The contribution is an audit framework, not a new domain-generalization method: the FSR ladder makes explicit what measurement is required to support what control conclusion, and the two boundary findings plus one honestly-blocked question show current observables do not certify shortcut reliance. The claim ledger (Table 1) and the route-inclusion ledger (Appendix B, Table 2) record every claim's status and every route's inclusion, so the boundary between what the evidence supports and what it does not is auditable.

A Provenance and reproducibility

All results re-analyse frozen artifacts from three internal repository lines; no model is trained, tuned, or selected on the analysed data. The CIGL graph-CMI line and the TOS erasure line are the internal evidence sources [Anonymous, 2026a,c]; the FBCSP-LGG branch line supplies the branch-load descriptives.

ID	Status	Claim
C1	READY_WITH_CAVEAT	Measured leakage magnitude does not certify reliance.
C2	READY_WITH_CAVEAT	Task-head alignment is closer to reliance than raw leakage in frozen CIGL R3.
C3	READY	Subject signal is erasable.
C4	READY	Erasure strength does not certify target benefit.
C5	READY_WITH_CAVEAT	Random-k falsifies nonspecific NLL movement.
C6	READY	Spatial branch is load-bearing.
C7	READY	Branch-local leakage/reliance is missing.
C8	READY	CMI-control remains closed.
C9	SUPPORT_ONLY	TTA-Control is positive but non-CMI.
C10	READY	FSR is an audit framework, not a new DG method.

Table 1: Claim ledger. Statuses are derived from the frozen result artifacts, not hand-set.

Artifact sources.

RQ	Pipeline	Frozen source (branch @ short-SHA)
RQ1/RQ3	CIGL graph-CMI (2a, 2015; LOSO; seeds 0,1,2)	cigl-r123-scaffold @ d3593c8
RQ2	TOS frozen-feature erasure (2a, Lee, Cho, HGD)	tos @ 1c65d79
RQ4	FBCSP-LGG branch ablation + gate	fbcsp-lgg-spatial-cmi-fusion @ 39c245a

Claim-strength tiers. Every quantitative statement carries one tier: `RECOMPUTED` (recomputed from committed per-unit data), `RECOMPUTED_SIGN_ONLY` (only a recomputable slice confirms the sign; e.g. per-fold graph leakage survives at seed0 only), `FROZEN_NOT_RECOMPUTABLE` (carried from a frozen output whose per-unit inputs were pruned — support only, never a reproduction). A fail-closed schema validator and a reproduction script (which halts if the headline drifts beyond tolerance) enforce these.

Target-label firewall. Target labels score the L6 endpoint only; they never enter probe/eraser fitting, subspace fitting, model selection, hyper-parameter selection, or early stopping.

Route family	fit uses target y?	eval uses target y?
CIGL (RQ1/RQ3)	NO (source-only)	YES (scoring)
TOS erasers (RQ2)	NO (source-only)	YES (deploy scoring)

Every route in the audit carries a fit-tag in {NO, YES_FORBIDDEN, AUDIT_ONLY, UNKNOWN}; only NO-tagged routes enter quantitative tests, and the single YES_FORBIDDEN route (a retracted, disclosed target-label leak in a legacy line) is excluded from all tests (Appendix B).

RQ4 checkpoint search (provenance of the blocked status). Answering RQ4 quantitatively requires per-branch leakage (L1) and per-branch reliance (L5), neither of which is on disk; more fundamentally, *no trained checkpoint exists*. A read-only search of the compute store (`/projects/.../yinghao`, maxdepth 6), the user home, and the git branches returned zero matching FBCSP-LGG checkpoints, consistent with a training pipeline that persists only scalar summaries and holds the best-epoch state in memory without serializing it. RQ4 is therefore reported as blocked; producing the metric would require retraining, which we do not do.

B Route inclusion ledger

The audit spans 37 routes across the repository's measurement/control lines. Under the Phase-1 quantitative-inclusion gate (Section 2), a route enters a cross-level test only if it carries ≥ 1 predictor-side level {L1–L4} *and* ≥ 1 endpoint-side level {L5,L6}. Six routes qualify; the remainder are support/boundary/protocol/background and inform interpretation only. No target-label-fit route enters any quantitative test.

Route	Predictor	Endpoint	Inclusion
CIGL	L1—L2—L4	L5—L6	INCLUDED
TOS_mean_scatter	L1—L3—L4	L6	INCLUDED
TOS_LEACE	L1—L3—L4	L6	INCLUDED
TOS_INLP	L1—L3—L4	L6	INCLUDED
TOS_RLACE	L1—L3—L4	L6	INCLUDED
TOS_random_k	L3	L6	INCLUDED
31 other routes: SUPPORT/BOUNDARY/PROTOCOL/BACKGROUND_ONLY (no target-label fit enters any RQ)			

Table 2: Route inclusion. Only routes with a predictor+endpoint pairing enter quantitative tests; no target-label-fit route does.

The excluded routes fall into four tags: SUPPORT_ONLY (a predictor or endpoint level is present but not the pairing a specific RQ needs — e.g. FBCSP branch load is L4-only), BOUNDARY_ONLY (a boundary/decision result such as the erasure refusal gate), PROTOCOL_ONLY (an executed protocol without an efficacy endpoint), and BACKGROUND_ONLY (context lines, e.g. the prior-decoupled adaptation decomposition). The full per-route ladder mapping, missing-metric policy, and target-label tags are released with the artifacts.

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